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# Urinary Cystine/Creatinine Concentrations Before and After Castration in Dogs With Suspected Androgen-Dependent Cystine Urolithiasis

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## ABSTRACT

**Background:** Androgen-dependent cystinuria has been described in intact male dogs. Castration has been recommended to reduce urinary cystine excretion.

**Hypothesis/Objectives:** Urinary cystine/creatinine concentration will significantly decrease in dogs with suspected androgen-dependent (Type III) cystinuria after castration and will be associated with a lack of sonographic evidence of urolithiasis.

**Animals:** Six intact male dogs with cystine uroliths.

**Methods:** Prospective, observational study. Urinary amino acid/creatinine concentrations and abdominal ultrasound were evaluated at baseline (T0), 30 days (T30), and 90 days (T90) after castration and urolith removal. No dietary change was recommended unless urolith recurrence was noted. DNA was evaluated for the cystinuria-associated marker for Type III cystinuria.

**Results:** The median decline in urinary cystine/creatinine (uCys/creat) concentration from T0 to T30 was 577 nmol/mg (IQR, 415–969; $p = 0.03$), from T0 to T90 was 798 nmol/mg (IQR, 580–989; $p = 0.03$) and from T30 to T90 was 34 nmol/mg (IQR, 19.5–443; $p = 0.03$). Two dogs had small (2.5–3 mm) uroliths at T30, which resolved in one of them at T90. Both dogs with ultrasonographic evidence of recurrence at T30 were positive for the cystinuria-associated genetic marker, although no signs of lower urinary tract disease were noted in either dog. Three of 6 enrolled dogs were homozygous positive for the genetic marker.

**Conclusions and Clinical Importance:** Castration should be considered for intact male dogs with cystine urolithiasis.

## 1 | Introduction

Cystine-containing uroliths comprised just under 3% of the total urolith submissionsfrom 2006 to 2018, but a significant non-linear increase in the proportion of cystine-containing uroliths was reported during that time ($p < 0.001$), from 1.4% (15/1100) in 2006 to 8.7% (49/565) in 2018 [1]. Cystine uroliths in dogs rarely contain other minerals, are generally of low

**Abbreviations:** uArg/creat, urinary arginine/creatinine concentration; uCys/creat, urinary cystine/creatinine concentration; uLys/creat, urinary lysine/creatinine concentration; USG, urine specific gravity.

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radiodensity, and are often not diagnosed until signs such as stranguria, pollakiuria, hematuria, or even urethral obstruction are noted.

Cystine urolith submissions from dogs have not only increased overall [1, 2], but they are more common in male dogs. Of the 273 male dogs from our laboratory with cystine-containing uroliths, 192 (192/273, 70.3%) were reported as intact (OR 16.7, 95% CI 12.5–20, $p < 0.001$) [1]. 88% of cystine uroliths were submitted from intact male dogs of various breeds to another laboratory [3]. These data suggest that being an intact male dog, regardless of breed, might be associated with Type III cystinuria.

While castration is recommended for male dogs with cystinuria to prevent genetic transmission, it is thought that castration reduces cystine excretion in dogs with Type III androgen-dependent mutations and might clinically "cure" this disease. This profoundly changes how cystine uroliths would be managed for some dogs and would allow clinical cure for a form of genetic urolithiasis, and long-term management, including dietary therapy, would not be required.

Cystinuria is considered a risk factor for cystine urolithiasis [4]. Retrospective data suggest that concentrations of urinary cystine, ornithine, lysine, and arginine (COLA) decrease after castration, usually in combination with a dietary change [5], prospective data regarding the effects of only castration on COLA excretion are lacking. Moreover, ultrasonographic urolith recurrence has not been reported after castration. For this study, we evaluated urinary amino acid concentrations in intact male dogs with cystine urolithiasis before and after castration and urolith removal. We hypothesized that urinary cystine/creatinine concentrations would decline to the reference range after castration. We also documented the presence or absence of clinical signs and sonographic evidence for uroliths.

## 2 | Materials and Methods

Intact male dogs with confirmed cystine urolithiasis were eligible for enrollment. Upon presentation, the primary reason for presentation to the hospital and the signalment were recorded, and a physical examination of the dog was performed. An abdominal ultrasonographic examination, lateral abdominal radiographs, CBC, serum biochemical panel, urinalysis with pH by meter, urine culture, and urine creatinine analysis were performed. Dogs were excluded if comorbidities were detected that would not allow them to complete the study or safely be anesthetized for the cystotomy and castration. Dogs with nephro- or ureterolithiasis that were causing ureteral obstruction were excluded. Urine for urine amino acid concentrations was collected during the day from all dogs, and food was withheld for approximately 10 h before collecting the urine specimens. Amino acid concentrations were determined using a Biochrom 30+ Amino Acid analyzer (Harvard Biosciences, MA). Homogenized urine specimens (40 μL) were prepared by mixing with 160 L of double distilled water and 200 μL of 6% sulfosalicylic acid, vortexed, and then cold-centrifuged at 12000 rpm for 15 min. Supernatant was collected and transferred to the analyzer. Urinary amino acid concentrations were compared to the laboratory's reference ranges. Blood was also obtained at day 0, and DNA was extracted and submitted to the PennGen Laboratories at the University of Pennsylvania for evaluation of the cystinuria-associated marker for Type III cystinuria.

After castration and urolith removal, focal urinary ultrasonographic examinations were performed to ensure complete removal of all uroliths, and dogs were discharged from the hospital when deemed stable by the attending clinician. Owners were instructed to continue feeding the dog's regular diet, including treats. Additional water consumption was encouraged. Dogs were reevaluated 30 days (T30) and 90 days (T90) after castration, and urine was collected again for urinary amino acid and creatinine concentrations, and a focal urinary ultrasound examination was performed. If urolith recurrence was noted at T30, defined by the presence of uroliths > 2 mm, the dog was transitioned to a commercial veterinary diet marketed for cystine prevention[1] but remained in the study. If no evidence of urolithiasis was noted at T30, diet change was not recommended.

## 3 | Statistical Analyses

Due to the data not being normally distributed, we used the nonparametric version of the two-sample *t*-test, the Wilcoxon Signed Rank Test, to test for differences across each time point, calculated as the value at the earlier time point minus the value at the later time point. The median and interquartile range (IQR) of the differences was reported. The comparisons were conducted using SAS software version 9.4 for Windows (SAS Institute Inc., Cary, NC). Any $p$ value < 0.05 was considered significant. As these were all planned comparisons, no adjustment for type I error inflation due to multiple testing was conducted.

## 4 | Results

### 4.1 | Dogs

Nine intact male dogs were screened, and six were confirmed to have cystine urolithiasis and were enrolled. There were two English Bulldogs, one French Bulldog, one Dachshund, one Pitbull terrier mix, and one Boxer mix. The median age was 5 years of age (range, 1–5). Three dogs presented with urethral obstructions, and the other three presented with lower urinary tract signs, including stranguria and hematuria. Routine closed castration was performed, and uroliths were removed via only cystotomy in five dogs and via cystotomy plus laser lithotripsy in one dog.

### 4.2 | Urinary Analytes

There was no statistically significant difference in the median urine specific gravity (USG) and urinary pH between any of the time points. Median USG was T0 = 1.025, T30 = 1.036, and T90 = 1.038. Median USG increased by 0.01 (IQR, −0.02–0.01) from T0 to T30 ($p = 0.31$), by 0.01 (IQR, 0.02 to −0.002) from T0 to T90 ($p = 0.09$), and by 0.004 (IQR, −0.01 to 0.001) from T30 to T90 ($p = 0.31$). Median urinary pH, measured by pH meter, was T0 = 6.5, T30 = 7.0, and T90 = 7.2. Urinary pH decreased by 0.22 (IQR, −0.49 to −0.13) from T0 to T30 ($p = 0.06$), by 0.32 (IQR,

−0.94 to 0.09) from T0 to T90 ($p = 0.44$), and increased by 0.01 (IQR, −0.18 to 0.07) from T30 to T90 ($p > 0.99$).

### 4.3 | Urinary Amino Acid Excretion

Reference ranges for the pertinent urinary amino acid/creat concentrations (nmol/mg) were: cystine: (9.2–426.6), ornithine (1.5–182.5), lysine (70.4–868.8) and arginine (6–85.8).

Urinary cystine, arginine, and lysine/creatinine concentrations (nmol/mg) at each time are depicted in Figures 1, 2, and 3, respectively. The median decline in urinary cystine/creatinine (uCys/creat) concentration from T0 to T30 was 577 nmol/mg (IQR, 414.7–969.3; $p = 0.03$), from T0 to T90 was 798 nmol/mg (IQR, 580.5–988.9; $p = 0.03$) and from T30 to T90 was 34 nmol/mg (IQR, 19.6–443.2; $p = 0.03$).

Median decrease in uArg/creat concentration (nmol/mg) between T0–T30 was 76.8 nmol/mg (IQR, 22.2–130.8; $p = 0.06$), from T0 = T90 was 61.9 nmol/mg (IQR, 25.7–139.7; $p = 0.03$) and between T30–T90 was 8.9 nmol/mg (IQR, 4.0–17.2; $p = 0.18$). Arginine concentration was below the lower limit of detection for one dog at T30.

The median decrease in uLys/creat concentration (nmol/mg) between T0 and T30 was 337 nmol/mg (112.2–1381.6; $p = 0.03$), between T0 and T90 was 407 nmol/mg (IQR, 180.8–1253.3; $p = 0.03$). The median increase in uLys/creat between T30 and T90 was 3 nmol/mg (IQR, −128.2 to 468.6; $p > 0.99$).

Urinary ornithine concentrations were below the detectable threshold at all three timepoints for three dogs, and for one dog at T0, and no statistical comparisons could be made among the dogs. Only two dogs had measurable urinary concentrations of ornithine at all three timepoints, and in only one dog was this above the reference range (one dog at T0 = 339 nmol/mg; reference range 1.5–182.5).

Ultrasonographic findings and the genetic test associated with Type III cystinuria are reported in Table S1 for all six dogs. While no dogs had lower urinary tract signs at any of the reevaluations, two dogs at T30 had small uroliths noted on ultrasonography. Three 2.5 mm uroliths were observed in one dog, and multiple uroliths (3 mm) were noted in the other. The diets of these two dogs were changed to the veterinary therapeutic diet with no other intervention. At the next assessment at T90, no uroliths were observed in one of the dogs, and only urinary "sand" was noted in the other. Only one of the dogs with ultrasonographic recurrence, but no lower urinary tract signs, at T30 were positive for the cystinuria-associated

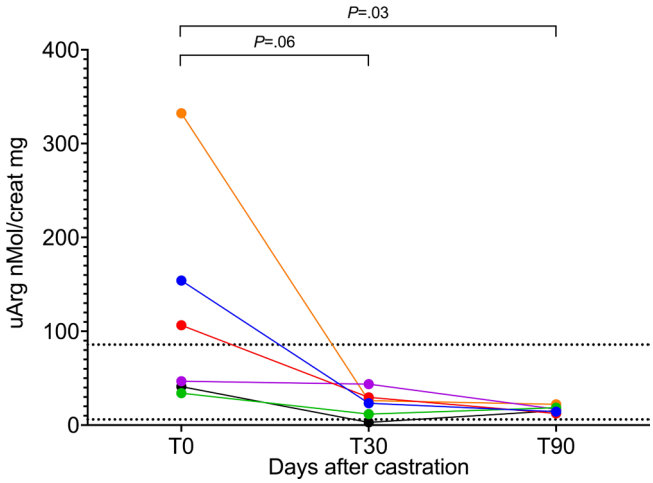


**FIGURE 2** | Urinary arginine/creatinine (uArg/creat) concentration at baseline (T0), 30 (T30), and 90 (T90) days after castration and cystolith removal in six dogs with suspected androgen dependent cystinuria. The dotted lines represent the upper and lower limits of the laboratory's reference range. Ultrasonographic urolith recurrence occurred in two dogs at T30 (orange and green lines) and urinary sand was noted at T90 in one dog (orange line).

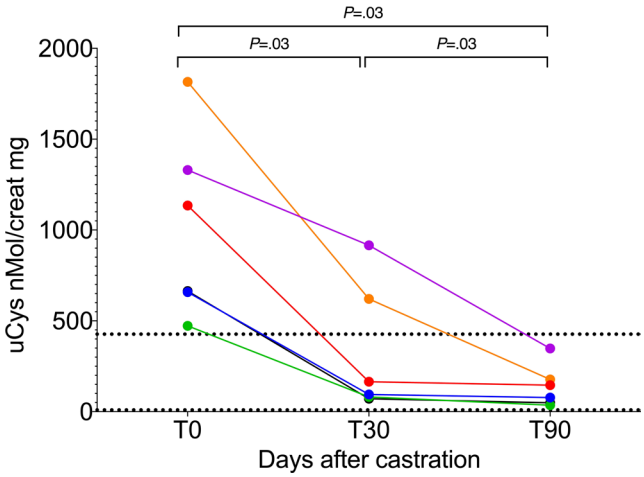


**FIGURE 1** | Urinary cystine/creatinine (uCys/creat) concentration at baseline (T0), 30 (T30), and 90 (T90) days after castration and cystolith removal in six dogs with suspected androgen dependent cystinuria. The dotted lines represent the upper and lower limits of the laboratory's reference range. Ultrasonographic urolith recurrence occurred in two dogs at T30 (orange and green lines) and urinary sand was noted at T90 in one dog (orange line).

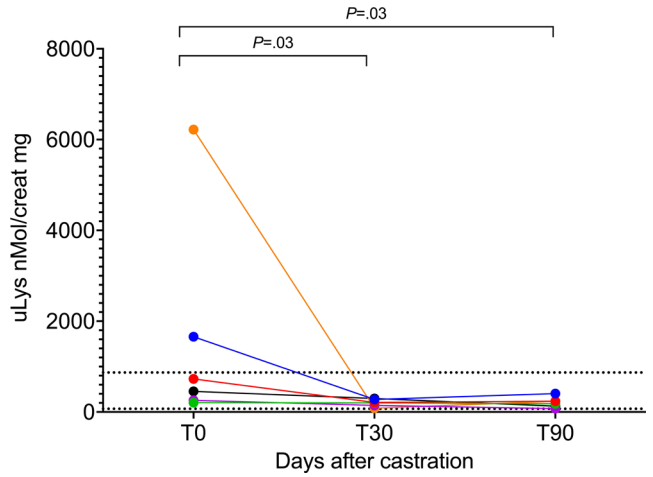


**FIGURE 3** | Urinary lysine/creatinine (uLys/creat) concentration at baseline (T0), 30 (T30), and 90 (T90) days after castration and cystolith removal in six dogs with suspected androgen dependent cystinuria. The dotted lines represent the upper and lower limits of the laboratory's reference range. Ultrasonographic urolith recurrence occurred in two dogs at T30 (orange and green lines) and urinary sand was noted at T90 in one dog (orange line).

genetic marker. Overall, 3/6 enrolled dogs were2–2 homozygous positive for the disease-associated marker (Boxer mix and both English bulldogs).

## 5 | Discussion

In this prospective study, we observed a significant decrease in uCys/creat concentration excretion 30 and 90 days after castration and urolith removal in six male dogs. Moreover, no dog had clinical urolithiasis during this time, despite two dogs having small cystoliths observed by ultrasonography on T30. At that timepoint, the diet of these two dogs was changed to a diet marketed for cystine urolithiasis prevention per study protocol. This was recommended because androgen-dependent cystinuria is likely not limited to specific breeds [3] and dogs with Type I and Type II cystinuria often develop recurrent uroliths. Both dogs with urolith recurrence at T30 had markedly decreased uCyst/creat compared to baseline (65% and 82% reduction for each dog, respectively), before dietary intervention. The dietary change could have optimized the urinary environment for urolith dissolution; however, no intervention might have led to the same result. A therapeutic diet marketed for preventing cystine urolithiasis might not be necessary if the uCyst/creat concentration significantly decreases from the concentrations noted before castration and there is no evidence of recurrent uroliths. Based on the USG values at T30 and T90, most owners were not consistently successfully achieving increased water intake. It is possible that any recurrence within 30 days of castration, and likely soon after surgery, might be prevented by urinary dilution.

The effects of dietary therapy should be further explored. Intact male dogs with cystine urolithiasis cannot be assumed to have the androgen-dependent form of the disease. Castration should always be recommended to reduce genetic transmission of the defect. Assessment of the current diet is always indicated, since reducing intakes of protein and sodium could help minimize urinary excretion of cystine [6]. A conservative approach should also include encouraging increased water intake to promote urinary dilution and reduce urinary cystine concentrations.

Our data are in agreement with a previous retrospective study that evaluated urinary COLA excretion before and after castration, where the mean uCyst/creat concentrations were markedly decreased after surgery (743.5 ± 629.6 umol/g before castration and 151.3 ± 182.5 umol/g creat after castration) [5]. However, in that study, two laboratories were utilized for amino acid analysis, and 24/37 (67%) dogs were already consuming a veterinary therapeutic diet before the first evaluation, while 83% were fed these diets after castration. Furthermore, the timing of the urine collections before and after castration was not provided, and no clinical or radiographic urolith recurrence rates were documented. In our study, the median percent decrease in uCyst/creat, uArg/creat, and uLsy/creat was 87.6%, 78.8%, and 64.8%, respectively, between T0 and T90. Ornithine excretion was below detection threshold in most dogs. While a larger sample size would increase the power of this small study, these preliminary data suggest that quantitative assessment of urinary cystine concentrations could be considered to monitor dogs with suspected Type III cystinuria after castration.

Median uCys/creat decreased by 84% (range 31%–89%) at T30 compared to baseline. Based on this limited information, clinicians might consider continuing dogs on their original diet (if otherwise appropriate) with water added and reevaluating 30 days after castration for urolith recurrence and urinary cystine concentrations. If a decrease in cystine excretion is not appreciated after castration or a pronounced urolith recurrence is noted, genetic testing for Type I and Type II cystinuria should be considered. Although testing for the cystinuria-associated marker is available for three breeds, androgen-dependent cystinuria likely occurs in many breeds and the exact genetic mutations remain unknown. Monitoring quantitative urinary cystine excretion before and after castration is recommended in lieu of, or in addition to, genetic testing for dogs suspected of having this disease. Furthermore, while dogs with Type I and II cystinuria often present at a young age (< 2 years of age), three of the dogs were 1 year of age. Therefore, the age and breed of the dog might not help elucidate the etiology for the cystinuria.

Limitations of this study include the small number of enrolled dogs, no long-term follow up, and each dog serving as its own control versus having a separate control group. In addition, the urolith itself could be the source of at least some urinary cystine. Finally, the possible effect of diurnal cycles on cystine excretion has not been explored in dogs and might have been a factor in our assessments.

In this small study, we were able to document significantly decreased urinary cystine excretion in dogs with suspected androgen-dependent cystinuria after castration and without diet change in most. Despite two of these dogs having nonclinical cystolith recurrence, no interventional procedure to remove the cystoliths was required. Measurement of urinary cystine/creatinine concentrations should be recommended ideally before and after castration in all intact male dogs with cystine urolithiasis to diagnose the androgen-dependent form of the disease. Moreover, castration is advised to help prevent recurrence and prevent genetic transmission of this disease.

**Acknowledgments**

The authors thank Dr. Urs Giger for his expertise in this area and consultation on these data.

**Disclosure**

Authors declare no off-label use of antimicrobials.

**Ethics Statement**

Approved by Institutional Animal Care and Use Committee at the University of California-Davis (protocol #22465). Authors declare human ethics approval was not needed.

**Conflicts of Interest**

Dr. Westropp is in the Purina Advisory Council and has collaborative grants with other companies. None of these are a conflict for the data presented in this study. Dr. Lasen also provides consultation and collaborations with pet food companies. This study was funded by internal funds and although two of the diagnostics (amino acid evaluation and genetic testing) are offered at the authors' institutions, no conflict was

present. None of the authors receive financial compensation if these tests are submitted to their respective institutions. The other authors declare no conflicts of interest.

## Endnotes

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## Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Ultrasonographic and cystinuria associated genetic marker results in six dogs with suspected androgen dependent cystinuria at baseline (T0), 30 (T30) and 90 (T90) days after castration and urolith removal.